Intro to Linear Transformations (Pt. 2)

Idea: Study linear maps that transform spaces of functions into new spaces of functions.

Given vector spaces V & W, a map $T: V \to W$ is linear (a "linear transformation") if

$$T(u + \alpha v) = T(u) + \alpha T(v) \quad \text{when} \quad u, v \in V.$$

Polynomial Interpolation

$$\begin{bmatrix} 1 & x_0 & x_0^2 & \cdots & x_0^n \\ 1 & x_1 & x_1^2 & \cdots & x_1^n \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^n \end{bmatrix} \begin{bmatrix} c_0 \\ c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \begin{bmatrix} f_0 \\ f_1 \\ f_2 \\ \vdots \\ f_n \end{bmatrix}$$

Given distinct points $x_0, \ldots, x_n \in [-1, 1]$, every continuous function $f: [-1,1] \to \mathbb{R}$ has a unique polynomial interpolant of deg $\leq n$:

$$p = P_x f$$

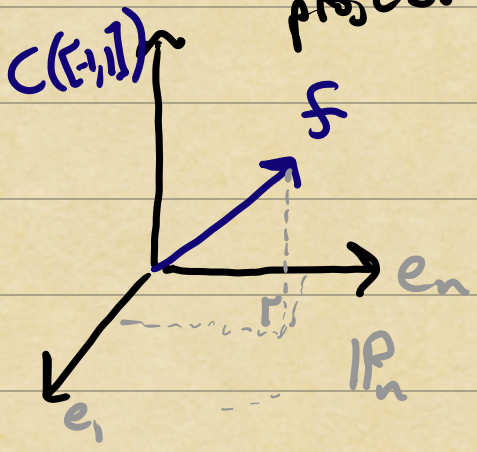
Fact 1: The map P_x is linear.

Fact 2: The map P_x is a projection $P_x^2 f = P_x f$

"Best"

$E_n = P_E f$

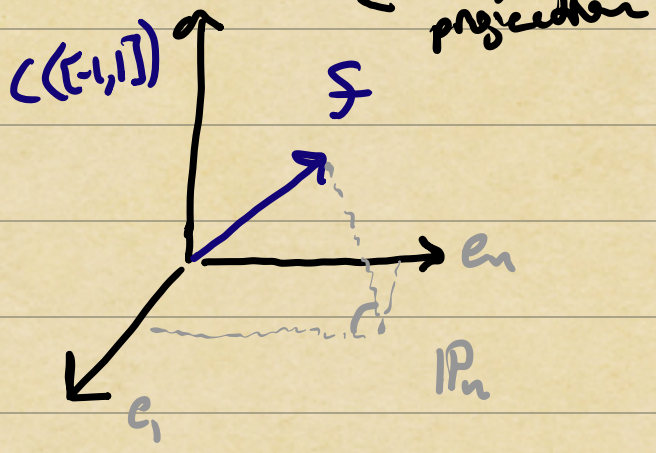
orthogonal projection



"Interpolant"

$E_n = P_x f$

oblique projection



The map $P_x : C[-1,1] \to P_n$ has operator norm

$$\|P_x\| = \sup_{f \in C[-1,1]} \frac{\|P_x f\|}{\|f\|} > 1$$

Claim: Given $X = \{x_0, \dots, x_n\} \subset [-1,1]$ and $E = \operatorname{span}\{1, x, \dots, x^N\}$, then

"Interpolation error" — "best-deg $\le n$ polynomial approx"

$$\|f - P_x f\| \le (1 + \|P_x\|)\, \|f - P_E f\|$$

$f - P_E f + P_E f - P_x f$

Pf $\|f - P_x f\| \le \|f - P_E f\| + \|P_E f - P_x f\|$

$\Rightarrow \quad P_x(P_E f) = P_E f$

$$\Rightarrow \|P_E f - P_X f\| = \|P_X(P_E f - f)\| \leq \|P_X\| \underbrace{\|P_E f - f\|}_{\|f - P_E f\|}$$

$$\therefore \|f - P_X f\| \leq (1 + \|P_X\|)\|f - P_n f\|$$

The "Matrix" of a Linear Transformation

If V and W have bases $V = \{v_1, \ldots, v_n\}$ and $W = \{w_1, \ldots, w_m\}$, then $T: V \to W$ has a matrix representation $T_{V \to W}: \mathbb{C}^n \to \mathbb{C}^m$.

$$T(a_1 v_1 + \cdots + a_n v_n) = a_1 T(v_1) + \cdots + a_n T(v_n)$$

$$= b_1 w_1 + \cdots + b_m w_m$$

$$\begin{bmatrix} T_{11} & \cdots & T_{1n} \\ \vdots & & \vdots \\ T_{m1} & \cdots & T_{mn} \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_n \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix}$$

$T_{V \to W}$ $\quad \underline{a}$ $\quad \underline{b}$

j^{th} column $\quad T(v_j) = T_{1j} w_1 + T_{2j} w_2 + \cdots + T_{mj} w_m$

Example: Differentiation $\frac{d}{dx} : \mathbb{P}_n \to \mathbb{P}_{n-1}$

$$\mathbb{P}_m = \{1, x, \dots, x^m\} \qquad m = n-1, n.$$

$$\frac{d}{dx} 1 = 0 \quad \frac{d}{dx} x = 1, \quad \dots, \quad \frac{d}{dx} x^m = m x^{m-1}$$

$$n \begin{bmatrix} 0 & 1 & 0 & & 0 \\ 0 & 0 & 2 & & 0 \\ 0 & 0 & 0 & \ddots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & m \end{bmatrix}$$

$n+1$

Idea: Choose a basis and reduce solving linear diff eq / integral eq. to matrix algebra.

Example: Find an approximate solution to

$$\frac{d}{dx} u(x) = u(x) \qquad \text{in } \mathbb{P}_n$$

$$\begin{bmatrix} 0 & 1 & & & \\ & 0 & 2 & & \\ & & \ddots & \ddots & \\ & & & & n \\ & & & & 0 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \\ a_n \end{bmatrix} = \begin{bmatrix} a_0 \\ a_1 \\ \vdots \\ a_{n-1} \end{bmatrix}$$

$a_1 = a_0$, $2a_2 = a_1$, $3a_3 = a_2$, ..., $n a_n = a_{n-1}$

recurrence relation

$a_2 = \frac{a_0}{2}$, $a_3 = \frac{a_0}{3 \cdot 2}$, ... $a_n = \frac{a_0}{n!}$

$$u(x) \approx a_0 \left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \cdots + \frac{1}{n!}x^n\right)$$

$n \to \infty$ $$u(x) = a_0 \sum_{j=0}^{\infty} \frac{x^j}{j!} = a_0 \exp(x)$$

↓ null space!

$$\left(\frac{d}{dx} - 1\right) u = 0$$

$$A \quad u = 0$$